# Effect of Marine and Atmospheric Heatwaves on Reflectance and Pigment Composition of Intertidal *Zostera noltei*

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**Abstract**

Seagrasses are critical to coastal ecosystems, providing habitat, stabilizing sediments, and aiding carbon sequestration. Climate change has increased the frequency and intensity of heatwaves, potentially threatening seagrass health. This study investigates the impact of marine and atmospheric heatwaves on the pigment composition and reflectance of the intertidal seagrass Zostera noltei. We performed laboratory experiments, exposing N. noltei samples to controlled heatwave conditions and measured hyperspectral reflectance and pigment concentration to assess its impact over time. Results revealed that heatwaves induce significant declines in seagrass reflectance, particularly in the green and near-infrared regions, linked to (likely due to) pigment degradation. Key vegetation indices, such as the Normalized Difference Vegetation Index (NDVI) and Green Leaf Index (GLI), also displayed marked reductions under heatwave stress, with NDVI values decreasing by up to 34% and GLI by 57%. A novel Seagrass Darkening Index (SDI) was developed to identify seagrass darkening, showing a strong correlation with heatwave exposure. This research suggests that spectral monitoring can effectively track the early impacts of heatwaves on seagrasses, providing a valuable tool for remote sensing-based habitat assessment. Satellite observations confirmed these findings, showing widespread seagrass darkening during atmospheric and marine heatwave events in Quiberon, France. Darkened seagrasses observed after heatwaves were exposed more than 13.5 hours daily.

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This work highlights the need for continuous monitoring of seagrass meadows under the current climate regime, underscoring the potential of remote sensing in capturing rapid environmental changes in intertidal zones.

*Keywords:* Remote Sensing, Pigment Composition, Seagrass, Coastal Ecosystems, Heatwaves

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## 1. Introduction

Intertidal seagrasses play a crucial role in the ecosystem by providing habitats and feeding grounds for various marine species, supporting rich marine biodiversity, and contributing significantly to primary production and carbon sequestration [1, 2]. These seagrasses are essential in maintaining the health of coastal ecosystems by stabilizing sediments, filtering water, and serving as indicators of environmental changes due to their sensitivity to water quality variations [3]. The interactions between seagrass meadows and their associated herbivores further enhance the delivery of ecosystem services, including coastal protection and fisheries support [4, 5, 6]. Understanding and preserving these ecosystems are vital for maintaining the biodiversity and productivity of coastal regions [7, 8].

Despite their crucial role in marine ecosystems, intertidal seagrasses face numerous threats that compromise their health and functionality. Coastal development and human activities are primary threats. These activities not only reduce the available habitat for seagrasses but also increase water turbidity, which limits light penetration and hampers photosynthesis [9]. Seagrasses are also threatened by nutrient enrichment from agricultural and urban runoff, which can lead to eutrophication. This condition promotes the overgrowth of algal blooms that compete with seagrasses for light and nutrients, further stressing these important plants [10] (Oiry et al. 2024). Pollution from industrial and agricultural fields sources introduces harmful chemicals and heavy metals into coastal waters, posing toxic risks to seagrass health. These pollutants can affect the physiological processes of seagrasses, reducing their growth and survival rates [11] Additionally, invasive species can out compete native seagrasses for resources, altering community structure and function [12].

Heatwaves, exacerbated by climate change, pose a growing threat to seagrasses. Marine Heatwaves (MHW), defined by [13] as prolonged discrete anomalously warm water events, and Atmospheric Heatwaves (AHW), defined by [14] as periods of at least three consecutive days with temperatures exceeding the 90th percentile, cause severe physiological stress on seagrasses [15, 16]. At the interface between the land and oceans, intertidal seagrasses are exposed to both MHW and AHW. Heatwaves have profound impacts on seagrasses, with their effects varying based on species and geographic location. For instance, the seagrass *Zostera marina* exhibits high susceptibility to elevated sea surface temperatures during winter and spring, leading to advanced flowering, high mortality rates, and reduced biomass [15]. Similarly, *Cymodocea nodosa* shows increased photosynthetic activity during heatwaves but suffers negative effects on photosynthetic performance and leaf biomass during recovery [16]. Additionally, different populations of *Zostera marina* along the European thermal gradient exhibit varied photophysiological responses during the recovery phase of heatwaves, indicating differential adaptation capabilities among populations [17]. These events intensify other stressors, such as overgrazing and seed burial, compromising sexual recruitment [18].

Remote sensing is increasingly being used to monitor marine ecosystems, including seagrass meadows. Spectral indices such as the Normalized Difference Vegetation Index (NDVI) and the Soil-Adjusted Vegetation Index (SAVI) are effective for quantifying vegetation health over time [19, 20, 21, 22]. By analyzing specific spectral patterns, remote sensing can track changes in seagrass meadows with high temporal and spatial resolution. Europe promotes these techniques through frameworks like the Water Framework Directive and the Marine Strategy Framework Directive, which advocate for habitat mapping using remote sensing technologies to cover large areas and detect long-term trends [23].

This study aims to experimentally test the hypothesis that warm events, such as atmospheric heatwaves, alter the pigment composition and reflectance of the intertidal seagrass Zostera noltei. By linking these changes with satellite remote sensing data, the study seeks to provide insights into how heatwaves affect seagrass meadows and to assess the potential of remote sensing in monitoring these impacts

## 2. Material & Methods

### *2.1. Laboratory Experiment*

#### *2.1.1. Sampling and acclimation of seagrasses*

Seagrass was sampled from a *Zostera noltei* (dwarf eelgrass) meadow on Noirmoutier Island, France (46°57'32.0"N, 2°10'37.0"W) at low tide in June 2024. A home-made inox sampling box was used to sample seagrass from an area of 30 cm by 15 cm and 5 cm deep, maintaining the sediment structure and avoiding damage to the rhizomes and the leafs of the seagrass (Figure 1 A). This sampling box allowed to limitate sampling variability between replicates. The entire system, including seagrass, sediment, meiofauna, and macrofauna, was placed in plastic trays together. Keeping the entire mesocosm allowed for the natural interactions between components to be maintained and reduced stress on the seagrass. To avoid hydric stress during transportation, seawater was added to each tray. Simultaneously, seawater was sampled from a nearby site and transported to the lab, where it was filtered using a 0.22 µm nitrocellulose filter to remove all suspended particulate matter. This seawater was used in the acclimation tank and the intertidal chambers. The seagrasses were acclimated at high tide for one weeks with a water temperature of 17°C, matching the temperature at the time of sampling, and with light of 150 µmol.s-1.m-2 of PAR photons [22]. A wave generator was used in the tank to circulate and reoxygenate the water.

Figure 1: Illustrations of the various steps of the experiment. A: Field sampling of seagrass using a homemade sampling box; B: Intertidal chambers used during the experiment; C: Seagrass sample inside a chamber during the experiment at high tide; D: Photo of the treatment sample at the start of the experiment; E: Photo of the treatment sample at the end of the experiment.

### *2.1.2. Experimental design*

Two intertidal chambers from ElectricBlue were used to simulate tidal cycles and control water temperature during high tide and air temperature during low tide (Figure 1 B,C). One chamber served as the control, while the other was used for the experimental treatment.

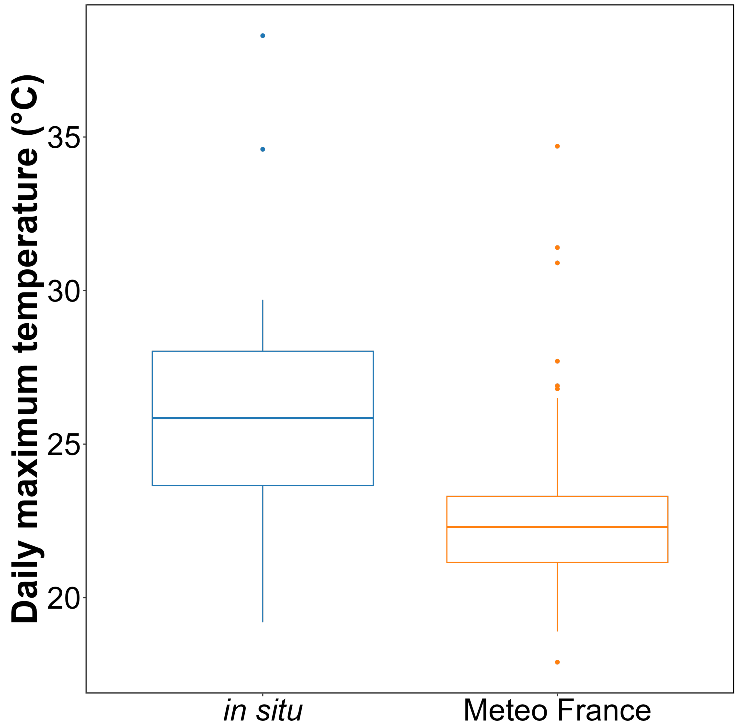


Figure 2: Comparison of daily maximum temperatures in August measured using an in-situ sensor (blue) and retrieved from Meteo France (orange). The solid line in the middle of the boxplot represents the median, the two ends of the box represent the 25th and 75th percentiles, and the whiskers represent values that are no more than 1.5 times the interquartile range.

To closely replicate field temperatures experienced by seagrasses during the experiment, *in situ* temperature sensors were installed at the sampling site in Bourgneuf Bay in August 2024. Daily maximum temperatures recorded by these *in situ* sensors were compared to measurements from the nearest Meteo France weather station (Figure 2). On average, *in situ* temperatures were 3 ± 3.2°C higher than those recorded by Meteo France. Additionally, temperatures recorded by Meteo France were more stable than those from the *in situ* sensors, likely due to the sheltered and shaded location of the Meteo France equipment. This difference was used to adjust heatwave temperatures measured by Meteo France to better reflect the conditions experienced by the seagrasses.

The control chamber was kept at temperatures representing typical seasonal conditions, with water temperatures ranging from 18°C to 19°C and air temperatures from 18°C to 23°C, following natural circadian temperature fluctuations (Figure 3 left). For the experimental treatment, air temperature was adjusted to replicate an atmospheric heatwave that affected the seagrass meadow in Quiberon, France (47°35'40.0"N, 3°07'30.0"W), from September 2 to September 6, 2021. On the first day of the experiment, air temperatures in the experimental chamber were set to range from 23°C at night to 35°C during the day, increasing by 1°C daily. Water temperature in the experimental chamber was also adjusted to mimic heatwave conditions, starting at the seasonal baseline (18°C) and rising incrementally by 0.5°C daily to simulate the increasing temperatures during the heatwave. This setup aimed to replicate the thermal stress experienced by the seagrass meadow during the actual heatwave event (Figure 3 right). The experiment concluded only when no changes in the reflectance of the treatment were observed for two consecutive days.

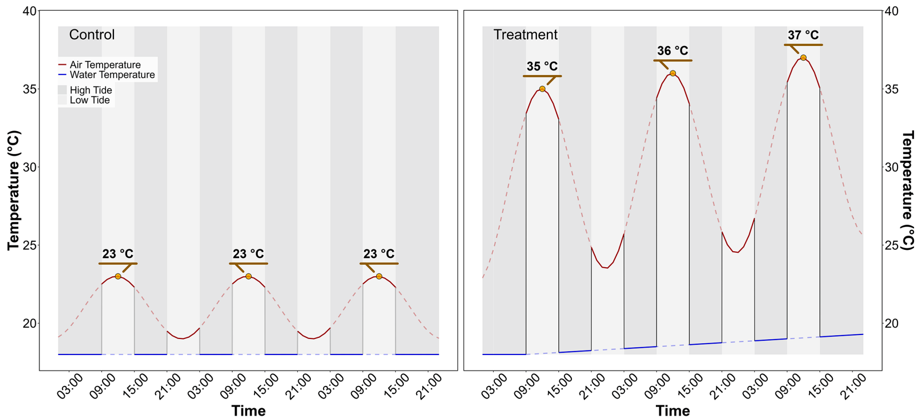


Figure 3: Temperature profiles of both the control (left) and the treatment (right) followed during the heatwave experiment. The red line indicates air temperature, whereas the blue line indicates water temperature. Due to the tidal cycle followed during the experiment, the seagrasses only experience temperatures representade by solid lines.

*2.1.3. Bio-optical measurmenents over seagrass leaves*

*2.1.3.1. Hyperspectral measurements.*

Throughout the experiment, hyperspectral signatures of both the control and treatment seagrasses were taken using an ASD HandHeld 2 equipped with a fiber optic, allowing measurements to be taken directly inside the chamber without opening it. Automatic spectra acquisition has been done using the RS3 softaware developed by the instrument manufacturer. An average of five reflectance spectrum ($R(\lambda)$), each with an integration time of 544 ms, was taken every minute. Every 10 minutes, the fiber optic was switched from one benthic chamber to the other, in order to measure reflectance in both treatment and control. Because light conditions were controlled inside of the chambers, reflectance calibration of the instrument was performed only each morning at the very first moment of low tide using a Spectralon white reference with 99% Lambertian reflectivity.

The second derivative of $R$ was calculated to retrieve absorption features and compare their variability over time. Two radiometric indices were also monitored throughout the experiment :

- The Normalized Difference Vegetation Index (NDVI, [24]), as a proxy of the concentration of chlorophyll-a (Equation 1)

$$NDVI = \frac{R(840) - R(668)}{R(840) + R(668)} \tag{1}$$

where $R(840)$ and $R(668)$ are the reflectance at 840 nm and 668 nm respectively.

- The Seagrass Percent Covers, developed by [25], used to assess proportion of seagrass inside of a pixel (Equation 2):

$$SPC = 172.06 \times NDVI - 22.18 \tag{2}$$

- The Green Leaf Index (GLI, [26]), as a measurement of the greenness of seagrass leafs (Equation 3)

$$GLI = \frac{[R(550) - R(668)] + [R(550) - R(450)]}{(2 \times R(550)) + R(668) + R(450)} \tag{3}$$

where $R(550)$ and $R(450)$ are the reflectance in green at 550 nm and in the blue at 450 nm, respectively.

*2.1.3.2. WILL BE REMOVED - Multispectral imagery measurement.*

Parallel to hyperspectral measurements, multispectral images were taken at the beginning and end of each diurnal low tide (09:00 am and 03:00 pm). A Micasense RedEdge-MX Dual multispectral camera, originally designed to be mounted on a drone, was modified for use without a drone. A 3D-printed mount was designed to attach the camera to the intertidal chamber and ensure that each picture was captured under the same conditions. At each time step (09:00 am and 03:00 pm), a first picture of the Spectralon was taken to allow for image correction in reflectance, followed by a second picture of the target. DISCOV, a Neural Network classification model previously developed to map intertidal vegetation using drone imagery, has been applied to each image taken inside the intertidal chambers. To understand the behavior of the model on seagrasses affected by heatwaves, classification images from before and after the heatwave have been compared.

#### *2.1.3.3. Pigment concentration measurements.*

At the beginning and the end of each diurnal low tide (09:00 am and 03:00 pm) leaves samples have been took in both the test and the control. leaves sampled have been stored under -80°C waiting for analysis. Before pigment extraction, each sample was lyophilized and its dry weight was measured. Pigment composition and biomass were analyzed using high-performance liquid chromatography (HPLC). The HPLC system (Alliance HPLC 248 System, Waters) was equipped with a reverse-phase C-18 separating column (SunFire C-18 Column, 100Å, 3.5 µm, 2.1 mm x 50 mm, Waters), preceded by a precolumn (VanGuard 3.9 mm x 5 mm, Waters). The system also included a photodiode array detector (2998 PDA) and a fluorimeter (Ex: 425 nm, Em: 655 nm; RF-20A, SH

**Au secours Philippe !!!**

### 2.2. Observation of seagrass leaves darkening.

#### 2.2.1. Field Observation

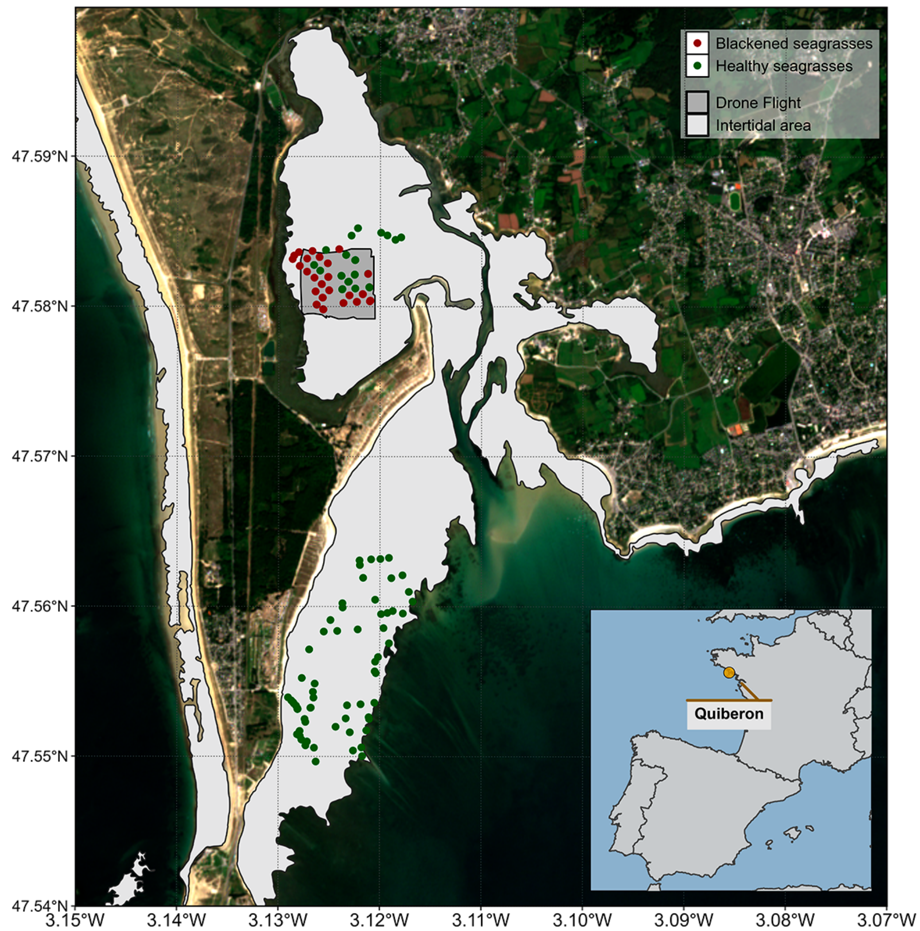


Figure 4: Location of the fieldtrip campaign that occured in the 10th of September 2021. The light grey polygons indicates the intertidal zone (Zone between High tide and low tide, that is totally emerged at low tide) and the dark grey polygon indicate the extent of the drone flight. Green dots indicate location of quadrat picture over green seagrasses while red dots indicate location of quadrat took over darkened seagrasses.

A fieldtrip, aiming to map a seagrass meadow near Quiberon (France : 46°57'32.0"N, 2°10'37.0"W), occurred in the 10th of September 2021 (Figure 4). During this fieldtrip, significant darkening of seagrass leaves was observed over large areas of the meadow (Figure 5 C & D). A total of 122 Ground Control Points (GCPs) were collected in the form of georeferenced quadrat images across the meadow. These images allow for visual assessment of vegetation type, density, and health status. The images were then divided into two categories:

green seagrasses and darkened seagrasses, based on a visual estimation of the leaf condition (Figure 4).

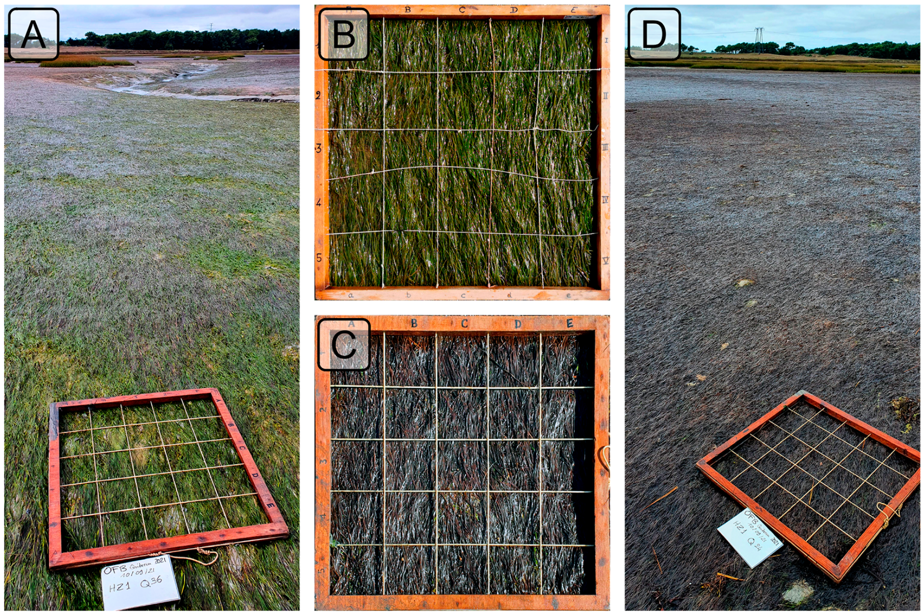

Figure 5: Illustrations of the two health conditions of seagrasses observed in the field. A: Global view of a green green meadow; B: Quadrat images of green seagrasses; C: Quadrat images of darkened seagrasses; D: Global view of an darkened meadow. All images were taken on September 10th, 2021, in Quiberon.

*2.2.2. Temparature Data and identification of heatwaves*
*2.2.2.1. Air temperature.*

Since January 1, 2024, Meteo France weather data has been freely and openly accessible. Hourly air temperature data (°C) for the French Atlantic and Channel coasts was retrieved using a custom script as no API was available at the time of this study. Weather stations located within 10 kilometers of the coastline were considered, but only those with at least 30 years of data were included to ensure reliable climatological reconstruction. Of the 156 weather stations within 10 kilometers of the coast, only 36 had sufficient data for climatology reconstruction. The hourly data was then aggregated into daily mean temperatures for each station.

Heatwave detection was performed using the HeatwaveR package in R [27]. This package utilizes the methodology proposed by [13] to detect heatwave events. The climatology for the year was computed using the temperature time series. An event was considered a heatwave each time the temperature exceeded the 90th percentile of the climatology for three consecutive days. The severity of each event has been assessed using the methodology proposed by [28]. Following the methodology of [29], coldspell events corresponding to temperature bellow the percentile 10th of the climatology during 3 consecutives days have also been measured.

Concerning heatwaves event near Quiberon, this methodology has been apply solely on the nearest weather station of our study site (Lorient-Lann Bihoue, 47°45'46"N 3°26'11"W , more than 395000 observation since the 1st of February 1952).

*2.2.2.2. Water temperature.*

SST data from the Copernicus CMEMS platform were downloaded for the French coast near Quiberon (47°29 03 N, 3°07 09 W), covering 1982-2022 [30]. Only pixels within an area of 2700 km² around Quiberon, Brittany, France (47°29 03 N, 3°07 09 W) were extracted and analyzed. This area was large enough to minimize missing values caused by cloud cover, yet small enough to avoid being influenced by the stability of offshore SST. The multi-sensor Level 3S Sea Surface Temperature (SST) product that we used is built from nighttime infrared satellite data, including sources like ESA SST CCI, C3S, and EUMETSAT. Data from AVHRR, ATSR, and SLSTR radiometers are used, but only those with a quality level above 4. An intercalibration method ensures consistency, and a daily reference SST field is constructed by combining the best data with median values from other sources. A large-scale bias correction is applied, and daily single-sensor composites are merged into a multi-sensor file (L3S), selecting the best sensor for each grid cell. A daily average was computed for the time series, and the HeatwaveR package [27] was used to compute SST climatology and detect SST events using the same method as air temperature event detection.

*2.2.3. Satellite Observation*

Two Sentinel-2 images covering the field trip site (Figure 4) were downloaded from the Copernicus platform [31]. The first image was taken before the heatwave event on September 1, 2021, and the second one after the event on September 6, 2021. Both images were acquired at Level-2 processing, in surface reflectance, and were orthorectified and atmospherically corrected to account for the effects of the atmosphere on reflectance values [32].

Reflectance value of each GCPs (Figure 4) have been exctracted and compared before and after the heatwave event.

*2.2.4. Emersion Time of Seagrasses*

To estimate the emersion time of seagrasses during low tide, bathymetric and tidal data were used. Bathymetry data for the Quiberon intertidal meadow were sourced from the "Service Hydrographique et Océanographique de la Marine" [33]. It corresponds to the Litto3D® product, which is a three-dimensional land-sea elevation database, accurately depicting the coastal terrain along certain regions of the French shores. Litto3D data were collected through airborne bathymetric lidar over the sea and airborne topographic lidar extending up to 2 km inland and is available as a gridded digital terrain model. it uses the NGF/IGN69 reference "zero", which corresponds to the mean sea level recorded at the Marseille tide gauge between 1885 and 1897, commonly known as the "Terrestrial Altimetric Zero".

Tidal data during the heatwave event were downloaded from the Intergovernmental Oceanographic Commission data portal [34], using measurements from the nearest tide gauge at Le Crouesty. In this dataset, the reference "zero" corresponds to the lowest astronomical tide, also known as the Hydrographic Zero.

Before calculating the emersion time, both datasets were intercalibrated to a common altitude reference. This involved applying a correction factor to the Litto3D data to align it with the Hydrographic zero. SHOM annually publishes a document called "Références Altimétriques Marines" [RAM, 35], which provides these correction factors for each seaport along the French coastline. For Le Crouesty port data from 2022, the correction factor was 2.85 m.

Once aligned, the corrected elevation was compared to water height for each pixel and each time step during the heatwave event. The emersion time was then calculated as the daily total time each pixel remained exposed.

## 3. Results

### *3.1. Laboratory Experiment*

#### *3.1.1. Spectral inspection*

Hyperspectral signatures were acquired throughout the experiment for both the Control and Treatment groups. In the Control group, reflectance remained relatively stable over time, with only minimal changes in magnitude and spectral features. Slight variations in spectral shape observed in this group were attributed to differing levels of water evaporation during low tides. Immediately after the water level receded from high to low tide, samples were very wet but dried progressively, reaching maximum dryness just before the tide rose again. This fluctuation in water content induced variations in the reflectance spectra. Overall, the Control group's reflectance spectra displayed a peak at 560 nm (green region), low values at 665 nm (indicative of strong chlorophyll-a absorption), and a high plateau in the near-infrared (NIR), beyond 705 nm (Figure 6, left).

In contrast, the Treatment group showed more pronounced changes in reflectance throughout the experiment. On the first day (T0), reflectance values were comparable to those in the Control group, especially in the visible region, with a notable peak around 560 nm and a pronounced trough at 665 nm. However, starting from T24 (the second day), reflectance began to decrease across all wavelengths, particularly around 560 nm and in the NIR. Around T48, the NIR reflectance appeared to stabilize at values like those observed at T24. In the green region, however, reflectance continued to decline slightly until the experiment's conclusion. Additionally, a slight increase in reflectance at 665 nm between T24 and T48 was observed, possibly associated with a reduction in chlorophyll-a concentration (Figure 6, right).

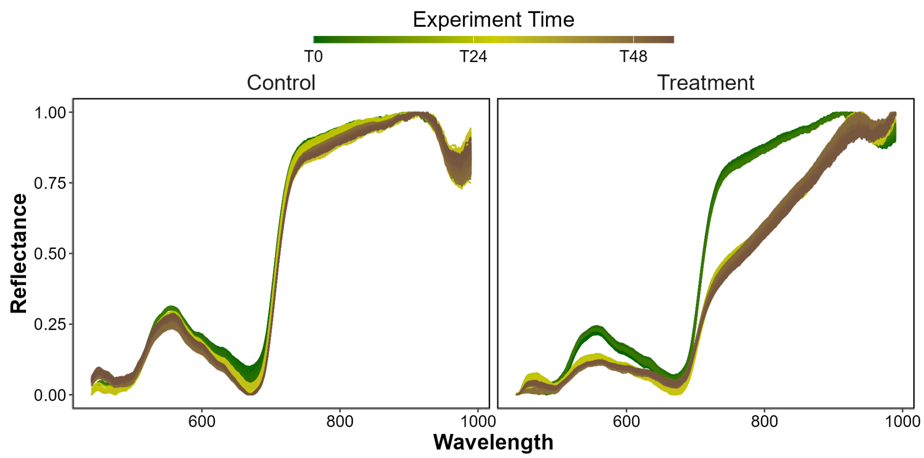


Figure 6: Hyperspectral Reflectance signature collected during the three days of the experiment for both the Control (Left) and the Treatment (Right). A min-max standardization has been applied to each individual spectrum. Temperature profiles applied to both scenarios are shown in Figure 3.

*3.1.2. Radiometric indices comparison*

The second derivative of the spectral signature for both Control and Treatment groups was computed, highlighting key absorption features. Figure Figure 7 A shows the average second derivative for the Control group (top) and the Treatment group (bottom) over the experiment's duration. Clear differences emerge between the two groups: the Control group exhibits more variability, reflecting more pronounced spectral features, whereas the Treatment group shows less variability, indicating fewer absorption features in its spectral signature. The maximum absorption for both groups occurs at 665 nm, with values of $1.63 \times 10^{-4}$ for the Control group and $8.08 \times 10^{-5}$ for the Treatment group—a reduction of 50.31% by the experiment's end.

Monitoring the second derivative at 665 nm ($R''_{665\,\mathrm{nm}}$) over time reveals a rapid drop in replicates 1 and 3, decreasing by 26 and 64 % respectively after 24 hours (T24) of treatment (Figure 7, B). After T24 $R''_{665\,\mathrm{nm}}$ in these replicates stabilizes for the remainder of the experiment. In contrast, replicate 2 shows a slower but more substantial decrease of 110 % by the experiment's end. Overall, between the start and the end of the experiment, the $R''_{665\,\mathrm{nm}}$ have decreased by 72 %.

NDVI shows the same trends, exhibit an initial rapid decline of approximately 25% between T24 and T30, after which their values stabilize, with minimal further changes observed through the end of the experiment. In contrast, Replicate 2 remains relatively stable between 0 and 48 hours, showing little change during this period (drop of 9% at T24). ,However, after T48, Replicate 2 undergoes a substantial drop, ultimately reaching a reduction of approximately 48% (Figure 7, C).

The relative GLI shows similar trends across the replicates. Replicates 1 and 3 exhibit an initial rapid decline of approximately 50 and 68 %, respectively between T24 and T30, after which their values stabilize with minimal further changes observed until the end of the experiment. In contrast, Replicate 2 remains relatively constant between 0 and 48 hours, showing only a minor decline of around 5% by T24. However, after T48 mark, Replicate 2 experiences a substantial decrease, ultimately reaching a reduction of 73% (Figure 7, D).

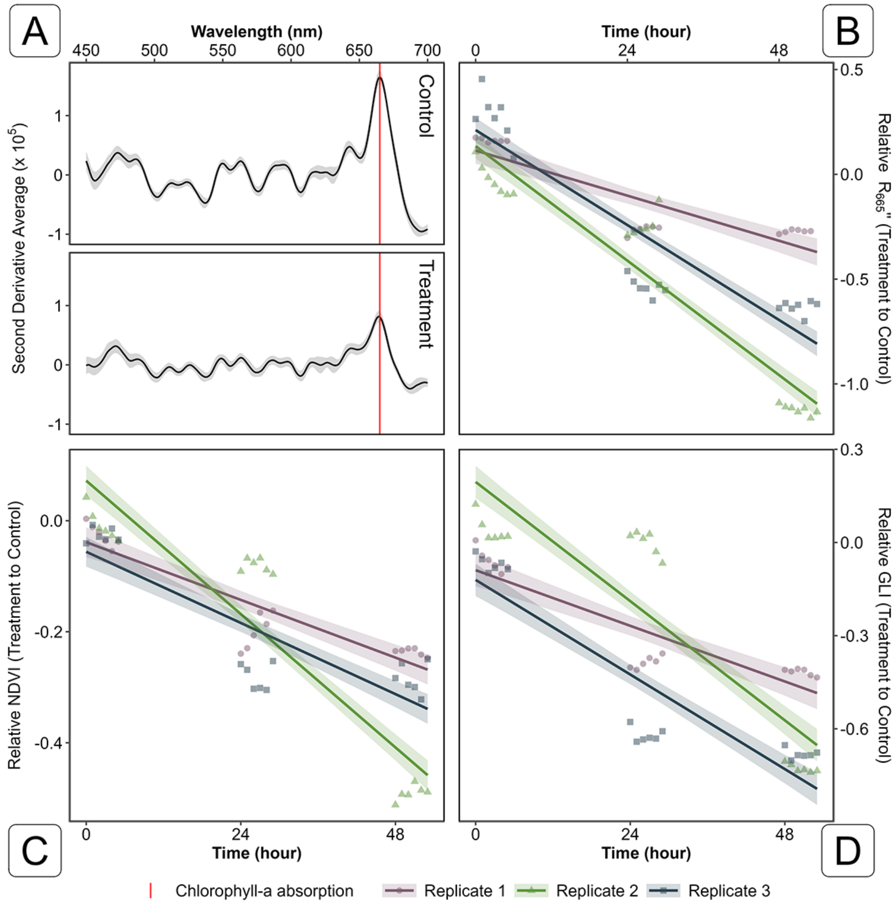


Figure 7: Second Derivative of the Control and the Treatment (A) and evolution of the second derivative (B), the NDVI (C) and the GLI (D) over Time. A: Mean and standard deviation of the second derivative for both Control and Treatment groups on the final day of the experiment; B: Relative evolution of the second derivative at 665 nm between Treatment and Control groups over time across three replicates. Points indicate raw data, the line represents a generalized linear model, and the shaded area the model's standard error; C: Relative evolution of NDVI for the Treatment group compared to the Control group over time across three replicates. Points indicate raw data, the line represents a generalized linear model, and the shaded area the model's standard error; D: Relative evolution of GLI for the Treatment group relative to the Control group over time across three replicates. Points indicate raw data, the line represents a generalized linear model, and the shaded area the model's standard error.

### *3.1.3. Buidling the Seagrass Darkening radiometric index*

Based on the observed spectral changes in seagrasses exposed to extreme events, a radiometric index can be developed to detect seagrass darkening events through remote sensing. This darkening is characterized by a reduction in reflectance within the green region, along with a substantial decrease in the rededge portion of the electromagnetic spectrum. (Figure 7).

The Seagrass Darkening Index (SDI) is introduced here as a measure of the difference between the interpolated value between 560 and 842 nm and the observed value at 740 nm. This approach calculates the line height using a triplet of bands, where 560 nm and 842 nm serve as the 'baseline' and 740 nm represents the central peak. The SDI can be mathematically expressed as:

$$\text{SDI} = [R(560) + 0.64 \cdot (R(842) - R(560))] - R(740) \tag{4}$$

Where $R(560)$, $R(740)$, and $R(842)$ represent reflectance values at 560, 740, and 842 nm, respectively. These wavelengths were selected to align with the spectral resolution of satellites like Sentinel-2, enabling the index to be used for mapping darkening events from satellite data. From SDI, negative values indicate green, healthy seagrasses, while positive values correspond to darkened seagrasses.

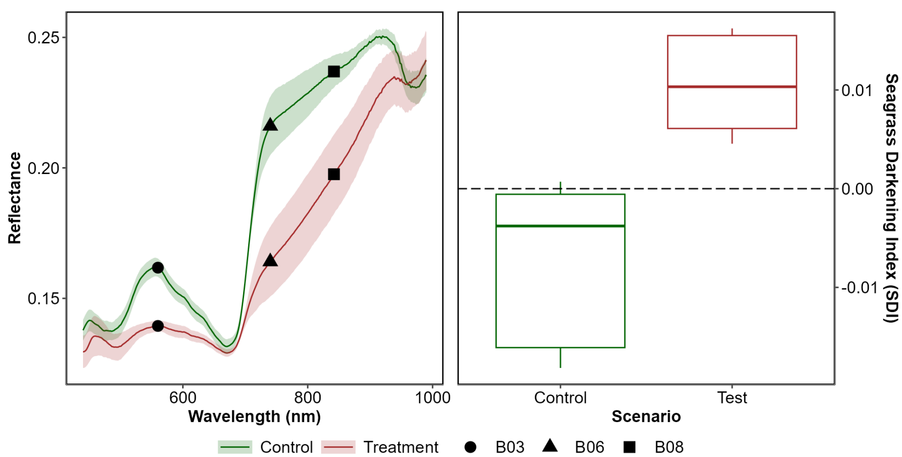


Figure 8: Development and Application of the Seagrass Darkening Index (SDI). The left panel displays the average reflectance values for both the Control and Treatment groups on the final day of the experiment. Shaded areas represent the standard deviation of reflectance, while markers indicate the spectral bands used to construct the SDI. The right panel shows the application of the SDI for both groups on the last day of the experiment.

When applied to reflectance data collected during the experiment, the SDI revealed clear distinctions between the Control and Treatment groups. By the end of the experiment, seagrasses in the Treatment group exhibited a median SDI of 0.01, with no samples showing a negative SDI value, consistent with their visibly darkened appearance. In contrast, the Control group retained a green appearance throughout the experiment, with a median SDI of -0.003, and only 3% of samples displaying a positive SDI value (Figure 8, right).

#### *3.1.4. Pigment composition evolution along the experiment*

bla bla bla bla

### *3.2. Heatwaves of Quiberon*

#### *3.2.1. Spectral inspection*

The Sentinel-2 images used in this study are dated from the 1st of September 2021 and the 6th of September 2021 (Figure 9 A and C, respectively). The Atmospheric Heat Wave (AHW) started on 4 September and lasted until 7 September, while the Marine Heat Wave (MHW) started on 3 September and ended on 8 September 2021 (Figure 9 B). The AHW has been classified as a strong event, and the MHW has been classified as moderate. Even though the Sentinel-2 of the 6th of September was captured only a few days after the start of the events, darkening of seagrass can already be observed in the true-color composition (Figure 9 C). Before the event, all GCPs appeared green un the images, with similar reflectance spectra, regardless of the class attributed to the GCPs on the field (Figure 9 A and D left). Their reflectance spectra showed a peak at 560 nm (in the green part of the spectra), low values at 665 nm, corresponding to the strong absorption by chlorophyll-a and a high infrared plateau (> 705 nm). However, GCPs classified as dark seagrass during the fieldtrip, showed significant differences in reflectance spectral shape compared to GCPs classified as green seagrass on the 10th of September, with corresponding differences in the true color composition (Figure 9 C and D right). In fact these dark seagrass GCPs were affected by extreme events, which altered their color and impacted the satellite reflectance (Figure 9 C). The reflectance spectra over dark seagrass were characterized by the loss of the reflectance peak at 560 nm and a decrease in the infared plateau, which was observed as a steady increasing slope up to 940 nm.

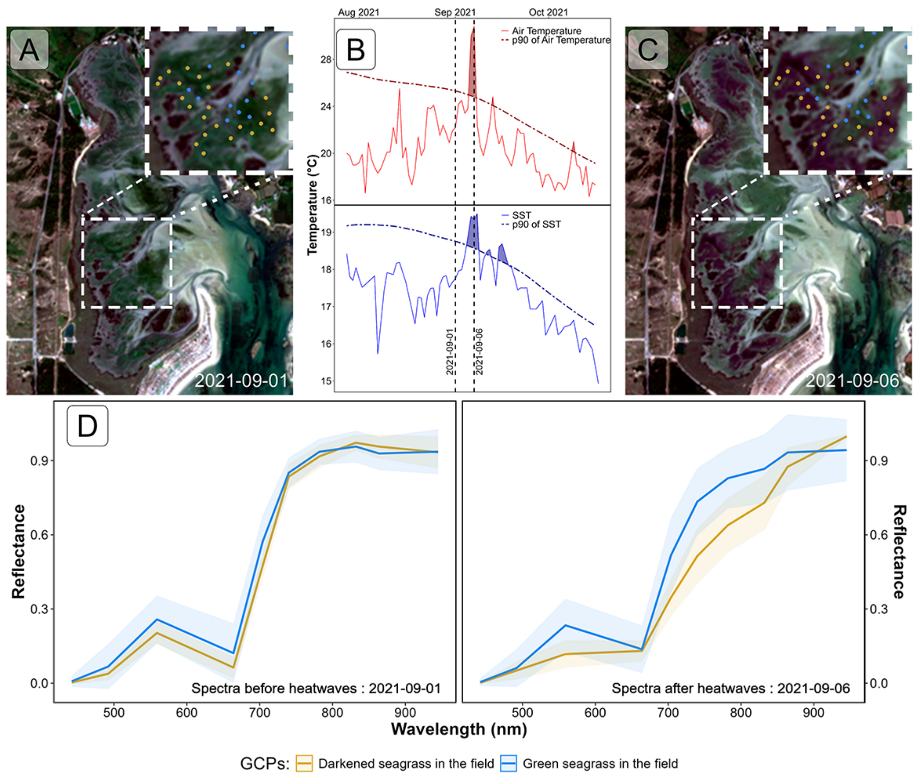


Figure 9: Spectral Comparison of Sentinel-2 Reflectance (D) over Ground Control Points (GCPs) Before (A) and During (B) Heatwave Events (C); A: RGB color composition of the Sentinel-2 image on the 1st of September 2021. The points correspond to GCPs collected on September 10, 2021 Figure 4; B: Detection of heatwave events based on both Air Temperature and Sea Surface Temperature (SST). The solid line represents the daily average temperature, while the dashed line indicates the 90th percentile of the climatology. Colored polygons identidy heatwaves (marine in blue and atmospherical in red). The two vertical dashed lines represent the acquisition dates of the two Sentinel-2 images used in this analysis ; C : RGB color composition of the Sentinel-2 image on the 6th of September 2021. The points correspond to GCPs collected on September 10, 2021 Figure 4 ; D : Average spectral signatures of GCPs where dark and green seagrasses (blue and gold lines, respectively) were identified during the field survey. The left plot shows the reflectance of these GCPs before the heatwave impact, while the right plot shows the spectral signature during the heatwaves. The ribbons around the lines represent the standard deviation.

### *3.2.2. Radiometric indices comparison*

Using Sentinel-2 data, the spectral indices (NDVI and GLI) and SPC estimated for pixels unaffected by the heatwave showed no significant differences before or during the event (Figure 10, left column) In contrast, seagrass darkened by the heatwave exhibited a significant decrease in both index values and SPC (Figure 10, right column). In this case, the NDVI dropped by approximately 34%, from a median of 0.61 to 0.40, while also becoming more homogeneous (the standard deviation decreased from 0.09 to 0.06) (Figure 10 B). Although the heatwave did not affect the observed seagrass density in the field, the drop in NDVI led to an underestimation of satellite-based SPC, from 83 to 48% (Figure 10 D). The GLI, the index that reflects the greenness of leaves, was the most affected by the event, with an important reduction from a median of 0.23 to 0.03 during the event (Figure 10 F). Interestingly, when comparing the initial seagrass density of both targets (i.e., seagrass density before the event) pixels affected by the heatwave (median SPC = 85%; Figure 10 D green) were denser than the those remained unchanged (median SPC = 65%; Figure 10 C green).

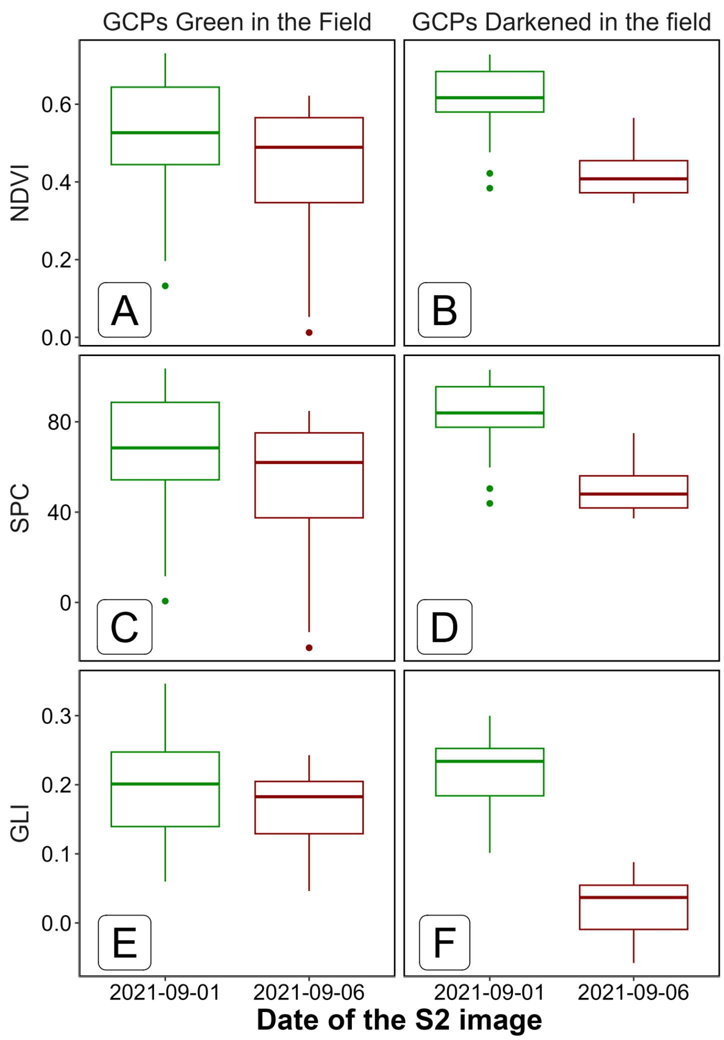


Figure 10: Comparaison of Sentinel-2 based radiometric indices at two different dates, before (2021-09-01, green) and after (2021-09-06, red) the heatwaves events of Quiberons, for the two category of GCPs seen in the filed : Green seagrasses (i.e seagrass not affected by the event, Left Column) and Darkened Seagrasses (i.e seagrass affected by the event, Right column). The Normalised Difference Vegetation Index (NDVI) has been computed using the Equation 1, the Seagrass Percent Cover (SPC) with the Equation 2 and the Green Leaf Index (GLI) using Equation 3,

### *3.2.3. Mapping of the seagrass darkening*

Using the Equation 4, we can detect seagrass patches that have darkened. Following the heatwave events, a total of 26.9 hectares of seagrass darkened, with the largest affected patch covering nearly 8 hectares. Overall, 25 % of the total seagrass meadow area showed signs of darkening between the 1st and

6th of September 2021. Comparing the spatial distribution of darkened patches with the site's bathymetry reveals that 94.6 % of darkened areas are located at above 3.8 meters of bathymetry (Figure 11, A and B).

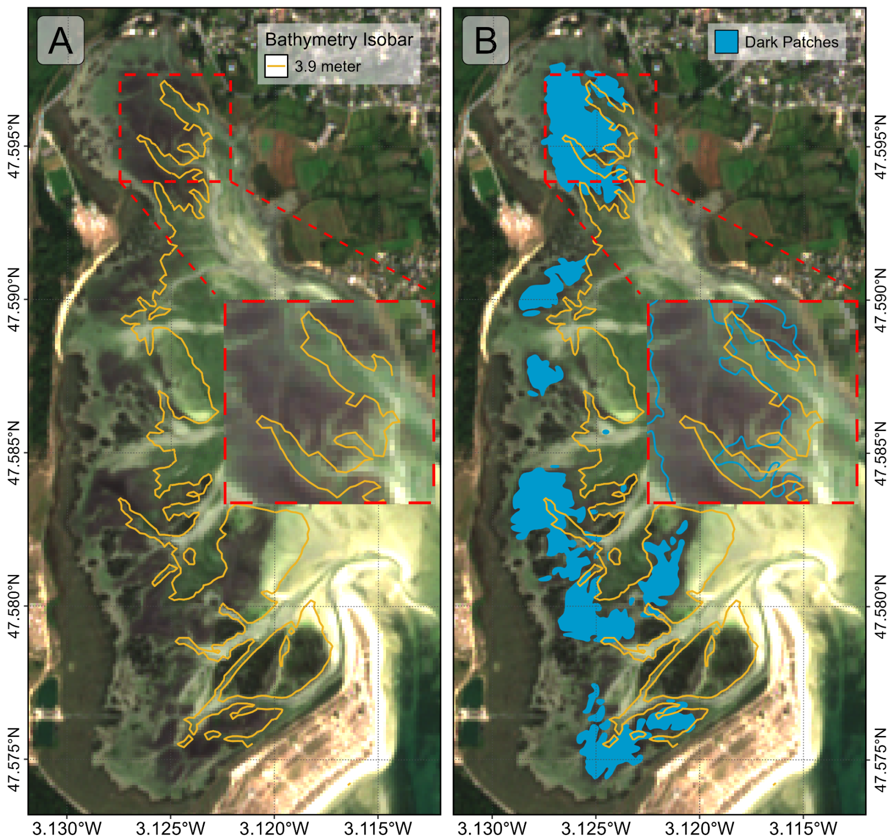


Figure 11: Map of the Darkening of seagrasses during the heatwave events of September 2021 in Quiberon. A: 3.9 m bathymetry isobar (golden line); B: Result of the detection of darkened seagrass patches (Blue plolygons)

Additionally, analysis of bathymetry data to determine seagrass emersion time shows a clear relationship between the time seagrass is exposed to air and its darkening. During the heatwave, no significant darkening occurred with daily exposure below 13.5 hours. However, when exposure reached 13.5 hours or more per day, seagrasses began to darken, peaking at around 14.5 hours of daily exposure (Figure 12).

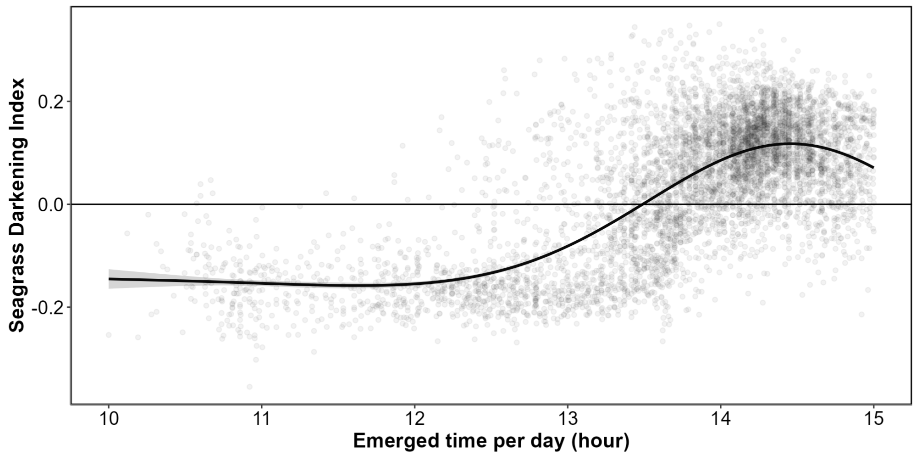


Figure 12: Seagrass Darkening Index as a function of the daily emergence time of seagrasses. The line represents a Generalized Additive Model (GAM) prediction, and the shaded area indicates the standard error. Shaded points represent raw data, with each point corresponding to a single pixel of the meadow.

## 4. Discussion - WORK IN PROGRESS

Although extensive research exists on marine heatwaves' effects on subtidal seagrasses, little attention has been given to intertidal regions and even less to the interaction between atmospherical extrem event and intertidal meadows. This study initiates an exploration of how intertidal seagrasses respond to the dual influence of marine and atmospheric heatwaves, underscoring the need for further investigation in this underexplored area.

In the present study, significant changes in the reflectance of seagrasses exposed to heatwaves were observed experimentally. These changes mainly include a drop in reflectance around 560 nm and another drop around 740 nm in the near-infrared part of the spectrum. The decrease in visible reflectance can be multifactorial and has been documented in plants under various stress conditions, including thermal stress. Leaf blackening in plants, as observed in *Protea neriifolia*, is primarily driven by oxidative processes involving the enzyme polyphenol oxidase (PPO) and the oxidation of phenolic compounds. When photosynthesis is inhibited by factors such as low light, chemical interference, or thermal stress, the plant's production of essential carbohydrates and antioxidants diminishes, increasing oxidative stress and leading to darkening. In the absence of photosynthesis, membrane integrity is compromised, allowing PPO to interact with phenolic compounds, thereby accelerating darkening. Low-oxygen environments and antioxidant treatments, like diphenylamine (DPA), have been effective in reducing these oxidative reactions. Furthermore, high temperatures can destabilize membranes, especially in chloroplasts, disrupting photosystem II and impairing recovery of photosynthetic function [36, 37]. Furethemore, has the chlorophyll-a is degraded, other pigments like carotenoids become more visible, renforcing the dark color of leaves. The destruction of membranes (Chloroplasts, thylakoids, cell walls) due to oxidative stress also contributes to the drop in reflectance in the Red Edge and near-infrared regions of the spectrum ($> 700$ nm). Infrared reflectance in plant leaves is high primarily due to the internal cellular structure that scatters light within the leaf. When sunlight enters the leaf, it is diffused and scattered through various layers, particularly the spongy mesophyll, which contains air spaces and cell walls with different refractive indices. This scattering effect is intensified because there is minimal absorption in the near-infrared range ($700 - 1300$ nm), allowing light to reflect back through the leaf surface. Once the structure of these walls is compromised, the infrared plateau decreases significantly [38].

**The unique reflectance properties of seagrasses under heat stress facilitate the detection of this stress through remote sensing techniques (Equation 4). The SDI index developed in this study allows for remote sensing assessment of the extent and severity of darkening events across intertidal seagrass meadows. Requiring only three spectral bands (560, 740, and 840 nm), this index can be applied to data from various current satellite missions, including Sentinel-2, Pleiades-Neo, WorldView-3, SkySat, and GeoSat-2 and will also be compatible with upcoming missions such as Sentinel-2 Next Generation and Landsat Next. As climate change progresses, the frequency and intensity of heatwave events are projected to increase, positioning remote sensing as an essential tool for monitoring cumulative ecological impacts on seagrass meadows. While satellite platforms provide extensive spatial coverage, limitations such as cloud interference may restrict the continuity of observations, particularly during peak heatwave periods. Integrating satellite-derived data with drone-based observations, which offer enhanced spatial resolution and flexible deployment, could address these limitations by filling temporal gaps in satellite revisit cycles. This combined approach enables precise, localized monitoring of seagrass meadows immediately following extreme events, thus capturing rapid physiological and structural changes that might otherwise remain undetected. Such datasets are critical for the early detection of ecosystem degradation, facilitating timely conservation interventions and enabling targeted management strategies. Moreover, continuous, multi-year data acquisition would bolster the development of predictive models to forecast future heatwave impacts, enhancing the capacity for adaptive management and long-term resilience planning for seagrass ecosystems.**

- make a link with the batymetry and modelisation of futur impact regarding heatwaves and sealevel rise

*4.1. Frequency and Intensity of Heatwaves: Implications for Seagrasses and Their Resilience.*

The rapid escalation of heatwave frequency and intensity globally is a defining characteristic of the current climate crisis, influenced heavily by anthropogenic activities and greenhouse gas emissions. Rising temperatures have led to the intensification of extreme weather events, particularly heatwaves, which are not only becoming more frequent but also lasting longer and reaching higher temperature peaks. Recent research suggests that the magnitude of these events has surged in the past decades, and projections indicate a continuation of this trend well into the future. For example, heatwaves that were once considered rare are now up to ten times more likely to occur, with some regions experiencing events three times as intense as those in the early 20th century [39, 40]. In our study site in Quiberon, a similar pattern emerges at the local scale. Data from a local weather station indicate an increasing trend in both the frequency and cumulative intensity of atmospheric heatwaves, while the same parameters for coldspells—defined as extreme low temperatures below the 10th percentile—show a marked decline in both frequency and intensity Figure 13.

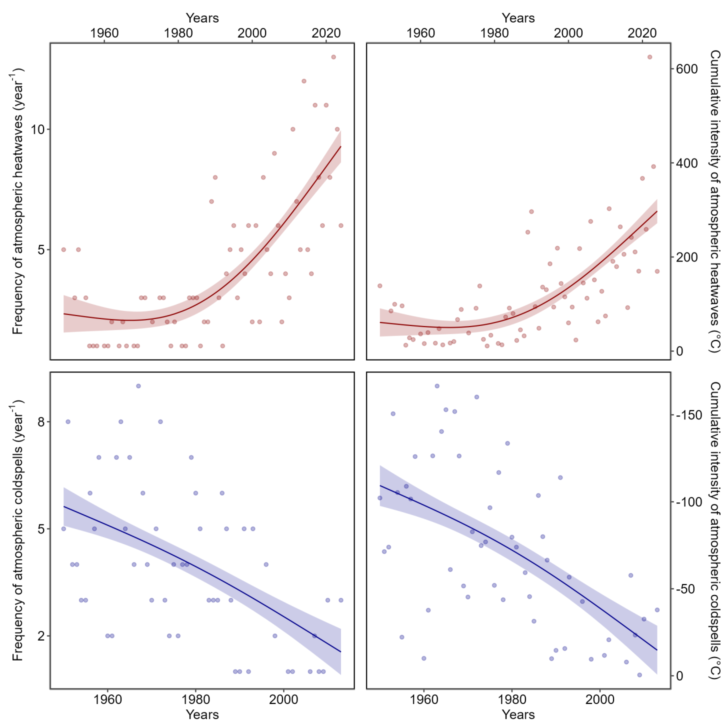


Figure 13: Frequency and intensity of atmospheric heatwaves (red) and cold spells (blue) over time at the Lorient-Lann Bihoué weather station. The left column displays the annual frequency of heatwaves and cold spells as the total number of events per year. The right column illustrates event intensity, represented by the cumulative temperature anomaly throughout the year, measured in degrees Celsius.

Metrics used to quantify heatwave intensity, such as cumulative indices based on total temperature exceedances over time, reveal a more comprehensive understanding of the severity and duration of these events compared to average temperature measures [40]. This is particularly relevant, as the cumulative impact of sustained high temperatures exerts more extensive stress on ecosystems, infrastructure, and human health than individual peak temperatures alone.

From a hazard analysis perspective, the concurrent evaluation of intensity, frequency, and duration of heatwaves allows for a nuanced understanding of their likely impacts. Models like the Heatwave Intensity Duration Frequency (HIDF) curves provide insights into the probability of various heatwave scenarios, indicating that both short but intense events and prolonged moderate events present unique risks depending on the region. In places such as the Mediterranean, for instance, the frequency of high-intensity heatwaves has dramatically increased, leading to severe impacts on urban infrastructure, public health, and energy consumption [41, 42]. Regions experiencing significant daily temperature variability are projected to see a sharper increase in extreme heat events, which underscores the role of atmospheric variability in shaping local heatwave dynamics. Specifically, in the mid-latitudes, where variability is expected to decrease, heat extremes may stabilize at elevated levels, thereby reducing cold extremes while allowing for increasingly frequent hot days [43].

In the ocean, similar trends are observed with marine heatwaves. These events are equally, if not more, consequential as atmospheric heatwaves, with widespread impacts on marine biodiversity, fisheries, and carbon cycling. MHWs have increased by over 50% in annual days since the early 20th century, and projections suggest that a near-permanent state of marine heatwaves could develop by the end of the century if greenhouse gas emissions remain high [44]. Events like the "Blob" in the northeast Pacific (2013–2016) have underscored the ecological ramifications of such heat anomalies, including shifts in species distributions, mass mortalities, and habitat degradation across vast oceanic regions. The physical mechanisms driving MHWs, such as altered ocean circulation and air-sea heat flux, further illustrate the interconnectedness of atmospheric and marine systems in the context of climate-driven thermal extremes [42].

The escalating frequency and intensity of heatwaves represent not only an atmospheric anomaly but a profound disruption to ecological stability across diverse ecosystems [39, 45]. The sustained high temperatures associated with both atmospheric and marine heatwaves lead to physiological stress, habitat degradation, and increased mortality in many species, particularly those with limited thermal tolerance [43, 44]. Terrestrial ecosystems experience shifts in species distributions, altered community dynamics, and reduced biodiversity as species are either forced to migrate or face local extinction under increasingly inhospitable conditions [46]. Similarly, in marine environments, prolonged heat stress from marine heatwaves has cascading effects on foundational species, including corals, kelps, and seagrasses, all of which are crucial for providing habitat, food, and shelter to diverse marine life [42, 44]. Seagrasses, in particular, play a vital role in carbon sequestration and coastal protection but are especially vulnerable to extreme heat events. Elevated temperatures can disrupt seagrass photosynthesis and metabolic processes, leading to reduced growth and heightened susceptibility to disease [18, 17, 15, 16]. With repeated heatwave exposure and limited recovery periods, seagrass meadows may suffer severe declines, threatening their ability to deliver key ecosystem services such as carbon storage, sediment stabilization, and habitat provision [41]. The compounded impacts of atmospheric and marine heatwaves thus pose an existential threat to seagrass ecosystems, highlighting the urgent need for targeted climate adaptation measures to mitigate these escalating thermal stresses and preserve the resilience of these essential marine habitats [45, 40].

Seagrass resilience to heatwaves is a complex and multifaceted issue shaped by species-specific traits, geographical location, and concurrent environmental stressors [47, 48, 49]. As climate change drives the frequency and intensity of marine heatwaves, understanding these dynamics becomes crucial, especially for temperate seagrass meadows composed of slow-growing, long-lived species like *Posidonia*, *Amphibolis*, and *Zostera*. These species, unlike their tropical counterparts, tend to be highly vulnerable to abrupt environmental changes, struggling to recover from disturbances due to their slower growth and longer lifespan. In contrast, colonizing species typical of tropical regions, such as *Halodule*, *Halophila*, and *Syringodium*, demonstrate greater resilience to warming events and marine heatwaves due to their rapid growth and more flexible life strategies [50].

The impact of heatwaves on seagrass ecosystems is further intensified by tidal variations, especially in temperate regions with tidal ranges that can exceed 10 meters, such as Mont Saint Michel Bay in France. This pronounced tidal amplitude affects intertidal seagrasses like *Zostera noltei*, which experience varying durations of air exposure based on their position within the intertidal zone. During neap tides, seagrasses situated higher in the intertidal zone may remain exposed for extended periods, heightening their vulnerability to atmospheric heatwaves. Conversely, seagrasses in the lower intertidal zone, although submerged longer, experience prolonged emersion during low tides, making them susceptible to extreme marine heat events. As a result, both marine and atmospheric heatwaves can increase stress on seagrass meadows, with effects that may be intensified by tidal timing and amplitude. In our study, we observed a darkening effect in the seagrasses, despite our experimental setup simulating tides with consistent 6-hour exposure periods. In natural settings, however, bathymetry plays a significant role, and depending on depth variations, seagrasses may be exposed for even longer durations during low tides. This suggests that *in situ* exposure times could exacerbate the stress effects observed, as seagrasses may endure prolonged air exposure beyond what our controlled conditions have replicated.

Species like *Zostera noltei* display distinct seasonal patterns that become increasingly pronounced at higher latitudes [51, 52]. These patterns reflect the species' adaptation to seasonal variations in temperature and light, but they also make N. noltei particularly sensitive to extreme events depending on their timing. For instance, if a heatwave coincides with early developmental stages or occurs after the biomass peak when leaf senescence has begun, the impact on meadow resilience can be severe

Environmental conditions can moderate seagrass resilience to thermal stress. Seagrass meadows located in areas benefiting from tidal cooling or positioned further from the warmer edges of their geographical ranges often experience reduced heat stress, resulting in higher shoot densities and enhanced resilience [47, 49]. In these cooler areas, *Nanostera noltei* exhibits high survival and photosynthetic performance up to 37°C, though temperatures above 39°C lead to near-total mortality within days, underscoring the species' sensitivity to temperature thresholds that may become increasingly common under climate change scenarios [53]. Moreover, the frequency, duration, and intervals between heatwaves significantly affect seagrass biomass and recovery; prolonged and frequent heatwaves reduce resilience and complicate recovery processes [48].

The influence of other stressors, such as eutrophication and sulfide accumulation, complicates this resilience. Increased sediment sulfide levels, which often accompany nutrient enrichment, can be toxic to seagrasses, particularly under elevated temperatures that amplify sulfide toxicity. *Zostera noltei*, for example, has a mutualistic relationship with lucinid clams that helps detoxify sulfides. However, this interaction is compromised at high temperatures, reducing the efficacy of sulfide removal and further inhibiting nutrient uptake, growth, and overall resilience [54].

In a simulated heatwave experiment on *Zostera noltei*, resilience was evident under short-term moderate stress, with no significant changes in photosynthetic performance or survival. However, prolonged or more intense heat events posed a greater challenge, highlighting the species' limited capacity to withstand chronic thermal stress [55, 53]. The long-term impact of heatwaves is especially evident in changes to seagrass cover (SPC). Before the heatwave event on 1st September 2021, impacted seagrass patches exhibited higher SPC than non-impacted areas. However, during the event on 6th September, SPC in the impacted patch experienced a sharp decline. By 7th October, one month post-event, the SPC of impacted seagrasses, initially higher, had fallen below that of non-impacted seagrasses Figure 14. The observed decrease in seagrass cover (SPC) on 6th September was primarily due to leaf darkening, which influenced the NDVI measurements used to estimate SPC. This darkening effect reduced the NDVI values, creating an apparent decrease in SPC in the impacted patch when, in reality, the density of seagrass remained stable. The bias introduced by remote sensing in this instance reflects a limitation in accurately capturing true seagrass cover during stress events. It was only after the heatwave that leaves began to detach, leading to an actual decline in seagrass density. This delayed physical response underscores how extreme events can compromise seagrass resilience, leaving previously robust patches in a weakened state compared to less disturbed areas Figure 14.

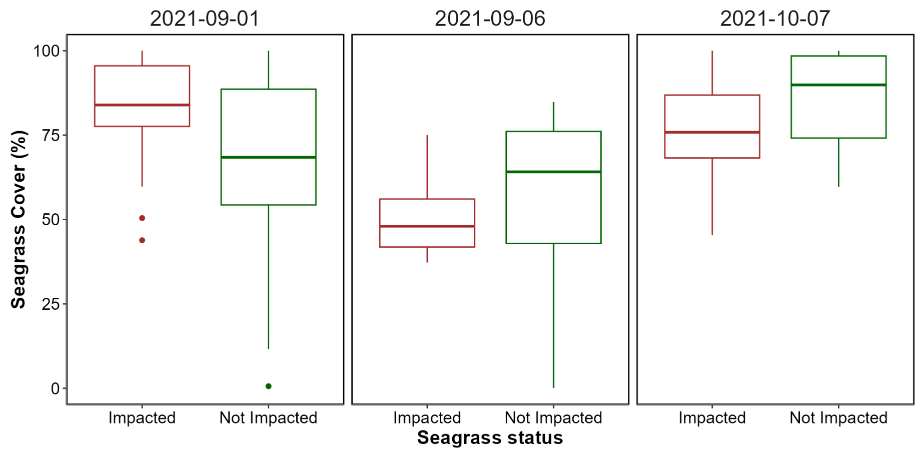


Figure 14: Comparison of the seagrass percent cover before (left), during (center) and after (right) the heatwaves for seagrasses that have not been impacted by heatwaves (i.e. remains green) and seagrasses that have been impacted by the events (i.e. seagrasses that darkened).

On a physiological level, seagrasses possess coping mechanisms such as photoprotective responses and heat-responsive gene expression, including the activation of heat-shock proteins (HSPs) and other stress-related genes [56, 57]. *Zostera noltei*, in particular, has shown differential gene expression responses to heat stress, with a variety of genes involved in protein folding, membrane stability, and reactive oxygen species scavenging playing critical roles. However, the rapid pace of climate change raises concerns about whether these adaptations can keep up with the increasing frequency and severity of thermal events [53].

Given the combined pressures of temperature extremes, eutrophication, and other anthropogenic impacts, targeted management strategies are essential for enhancing seagrass resilience [58]. Approaches such as reducing local stressors, cultivating heat-tolerant genotypes, and investing in restoration initiatives are vital to supporting these ecosystems in a warming climate. Although challenges remain, the adaptability and potential resilience of certain seagrass species offer hope for their persistence amid accelerating ecological shifts. In particular, regions that can buffer seagrasses from extreme stressors or provide cooler refuges may play a critical role in maintaining these valuable ecosystems in the face of global climate change [54, 47].

**Seagrass meadows function as foundational components of coastal ecosystems, sustaining diverse marine communities by providing essential habitats, nursery grounds, and trophic resources for fish, invertebrates, and other organisms. Recurrent heatwave events that damage these meadows trigger cascading ecological effects that extend beyond the direct impacts on seagrasses. Specifically, leaf darkening and subsequent declines in seagrass density compromise the**

**meadow's role in sediment stabilization and wave attenuation, which may heighten risks of coastal erosion.**

**From a biodiversity perspective, the degradation of healthy seagrass beds disrupts coastal food webs. Species reliant on seagrass habitats, including commercially valuable fish and shellfish, experience reduced access to food and shelter, potentially leading to declines in their populations. This effect has direct repercussions on local fisheries, impacting coastal communities whose livelihoods depend on these resources. Furthermore, as seagrass meadows decline, their capacity for carbon sequestration diminishes, indirectly contributing to elevated atmospheric carbon levels and exacerbating climate change impacts.**

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